

FLIN FLON, MB, Canada

WMO: 719540

Lat: 54.680N

Lon: 101.680W

Elev: 304

StdP: 97.73

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.7	-33.1	-39.8	0.1	-35.4	-37.2	0.1	-32.8	8.5	-16.5	7.8	-17.2	1.3	50	0.654

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	27.9	18.2	26.2	17.4	24.6	16.6	19.9	25.2	18.8	23.9	17.9	22.6	4.0	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.0	13.4	22.3	17.0	12.6	21.1	16.0	11.8	20.2	58.5	25.2	54.9	24.0	51.8	22.6	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.3	6.4	DB	-39.1	31.6	2.9	1.3	-41.2	32.5	-42.9	33.3	-44.5	34.1	-46.7	35.1	(4)
(5)			WB	-39.3	22.3	2.9	1.7	-41.4	23.5	-43.1	24.5	-44.7	25.4	-46.8	26.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.3	-19.7	-17.1	-9.7	0.3	8.2	15.2	18.5	17.1	11.2	3.0	-7.7	-16.5	(6)	
	DBStd	14.73	8.05	7.46	8.03	6.41	4.88	3.77	3.06	3.35	4.46	4.48	6.71	7.98	(7)	
	HDD10.0	4293	919	760	611	293	93	5	0	1	38	220	531	823	(8)	
	HDD18.3	6654	1178	993	869	541	314	106	36	63	217	474	781	1081	(9)	
	CDD10.0	761	0	0	0	2	37	161	263	221	74	4	0	0	(10)	
	CDD18.3	81	0	0	0	0	1	12	41	25	3	0	0	0	(11)	
	CDH23.3	632	0	0	0	0	21	110	300	177	23	0	0	0	(12)	
	CDH26.7	109	0	0	0	0	3	19	52	32	3	0	0	0	(13)	
(14) Wind	WSAvg	2.8	2.5	2.6	2.7	2.8	2.9	2.9	2.9	3.0	3.1	3.2	3.0	2.5	(14)	
(15) Precipitation	PrecAvg	469	18	16	17	26	37	67	83	61	62	36	25	22	(15)	
	PrecMax	664	35	33	33	49	81	124	130	140	103	76	75	47	(16)	
	PrecMin	384	6	1	2	2	4	22	39	25	12	4	10	2	(17)	
	PrecStd	77	9	8	10	13	20	27	25	36	28	16	17	11	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.1	3.7	10.0	19.4	26.0	29.0	30.1	29.8	25.9	17.6	7.8	1.2	(19)	
		MCWB	0.8	1.0	4.6	10.0	14.1	17.7	19.5	19.5	17.0	11.2	4.1	-0.1	(20)	
	2%	DB	-2.0	0.3	7.1	15.5	22.5	26.1	28.1	27.4	23.1	14.0	4.8	-1.8	(21)	
		MCWB	-2.9	-0.9	3.0	8.3	12.6	16.3	18.4	18.4	15.9	9.8	2.6	-2.6	(22)	
	5%	DB	-4.9	-2.5	4.0	12.4	20.0	24.2	26.4	25.3	20.4	11.6	2.7	-3.7	(23)	
		MCWB	-5.4	-3.4	1.2	6.4	11.5	15.4	17.8	17.6	14.5	8.0	1.0	-4.3	(24)	
	10%	DB	-7.4	-5.6	1.7	9.7	17.3	22.4	24.9	23.3	18.2	9.4	0.9	-5.5	(25)	
		MCWB	-7.9	-6.4	-0.1	4.5	10.0	14.6	17.1	16.7	13.2	6.6	-0.2	-6.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.0	1.4	5.0	10.8	15.7	20.2	21.6	21.1	18.5	12.5	4.8	-0.1	(27)	
		MCDB	1.9	3.1	9.7	18.0	23.6	25.4	27.0	26.9	23.4	16.0	6.6	1.1	(28)	
	2%	WB	-2.9	-1.0	3.0	8.5	13.5	18.4	20.1	19.7	16.6	9.8	2.6	-2.5	(29)	
		MCDB	-2.1	0.2	6.6	14.8	20.3	23.1	25.3	25.0	21.8	13.0	4.4	-1.8	(30)	
	5%	WB	-5.4	-3.5	1.6	6.7	12.1	17.0	19.1	18.6	15.1	8.4	1.2	-4.2	(31)	
		MCDB	-4.9	-2.4	3.9	11.8	18.4	21.8	24.5	23.4	19.9	11.3	2.6	-3.6	(32)	
	10%	WB	-7.8	-6.3	-0.1	4.8	10.6	15.7	18.1	17.6	13.7	6.9	-0.1	-6.0	(33)	
		MCDB	-7.3	-5.6	1.6	9.3	16.4	20.5	23.1	22.0	17.6	9.1	0.9	-5.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.9	11.3	12.8	12.1	11.5	10.3	9.9	9.2	8.8	6.5	6.5	7.9	(35)	
		MCDBR	10.2	11.9	13.4	15.2	16.1	13.4	12.3	12.3	12.4	10.4	7.6	9.2	(36)	
	5% WB	MCWBR	9.7	10.4	9.7	9.1	7.9	5.8	5.3	5.6	6.8	6.5	5.8	8.7	(37)	
		MCWBR	10.0	11.4	12.4	14.8	14.0	11.0	11.0	10.4	11.6	9.7	7.1	9.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.237	0.247	0.270	0.329	0.339	0.368	0.390	0.356	0.318	0.289	0.248	0.221	(40)	
		taud	2.300	2.333	2.356	2.387	2.460	2.401	2.385	2.481	2.577	2.586	2.441	2.306	(41)	
	Ebn at Noon		742	866	919	899	907	879	850	863	860	815	733	688	(42)	
		Edh at Noon	53	73	92	104	102	110	110	94	75	59	46	42	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.87	1.97	3.50	5.00	5.52	5.57	5.58	4.57	3.12	1.70	0.79	0.56	(44)	
	RadStd		0.09	0.14	0.30	0.34	0.57	0.44	0.36	0.35	0.36	0.24	0.09	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	+1.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		N/A	N/A	+1.60	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page